



Immune Profiling Reveals Distinct Features of Multiple Myeloma Subtypes Defined By Multi-Omics Network Analysis.



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Sinai

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BACKGROUND

- Multiple myeloma (MM) is characterized by a highly heterogeneous genomic landscape, posing significant challenges for patient stratification and management.
- Traditional classification systems often overlook the complex interactions between myeloma cells and the tumor microenvironment (MM-TME), which can influence disease progression and response to therapy.
- In this study, we complemented our bulk tumor patient classification MM-PSN (Patient Similarity Network) (Bhalla et al, Sci Adv 2021) with TME profiling using data from the recent MMRF Immune Atlas (Pilcher et al., BioRxiv 2024) to optimize patient stratification and uncover potential therapeutic targets based on the MM / MM-TME.

METHODS

Patient selection and study design

- We stratified 185 newly-diagnosed MM (NDMM) patients from MMRF CoMMpass, included in both our previous MM-PSN model and the MMRF Immune Atlas, into the following groups: (i) hyperdiploidy (HD) and tMYC [t(8;14)] (**MM group 1**, n = 106); (ii) tMMSET [t(4;14)] and tMAF [t(14;16)] (**MM group 2**, n = 46); (iii) tCCND1 [t(11;14)] (**MM group 3**, n = 33).

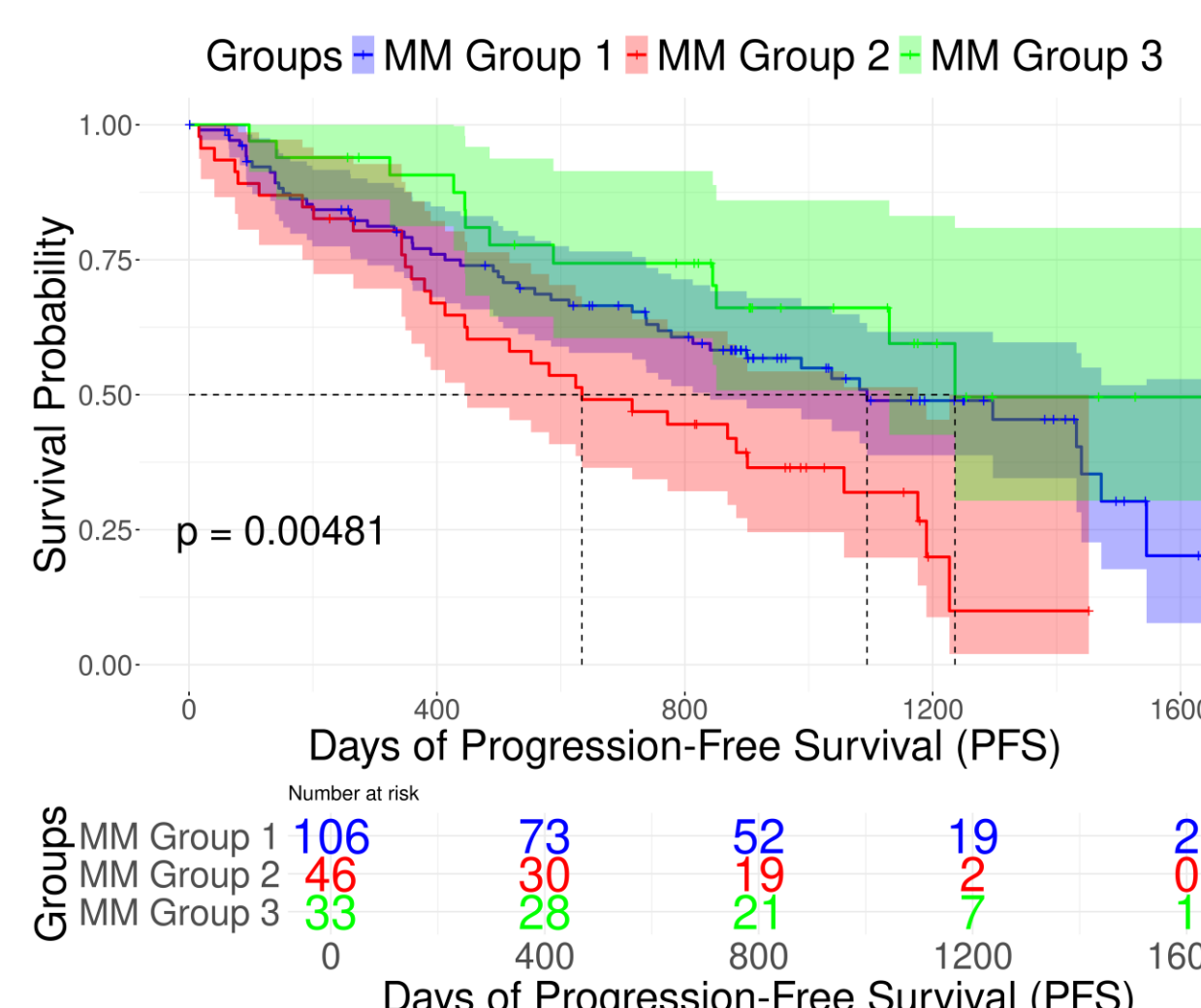
Immunophenotyping by 3' single cell RNA-seq

- Single cell RNA sequencing was performed to understand transcriptional changes in the CD138- fraction of BM aspirates from 185 samples collected at diagnosis.
- Samples were filtered based on total cell count (> 1000 cells), total unique molecular identifiers (UMIs) count (> 500 UMIs) and mitochondrial gene expression.
- Our dataset was comprised of 776,859 BM cells, subdivided in 85,147 myeloma cells, 107,884 myeloid cells, 407,895 NK / T cells, 162,630 erythroid and B cells
- Downstream analysis was performed using R packages Seurat, CellphoneDB, SCENIC and MOVICS.

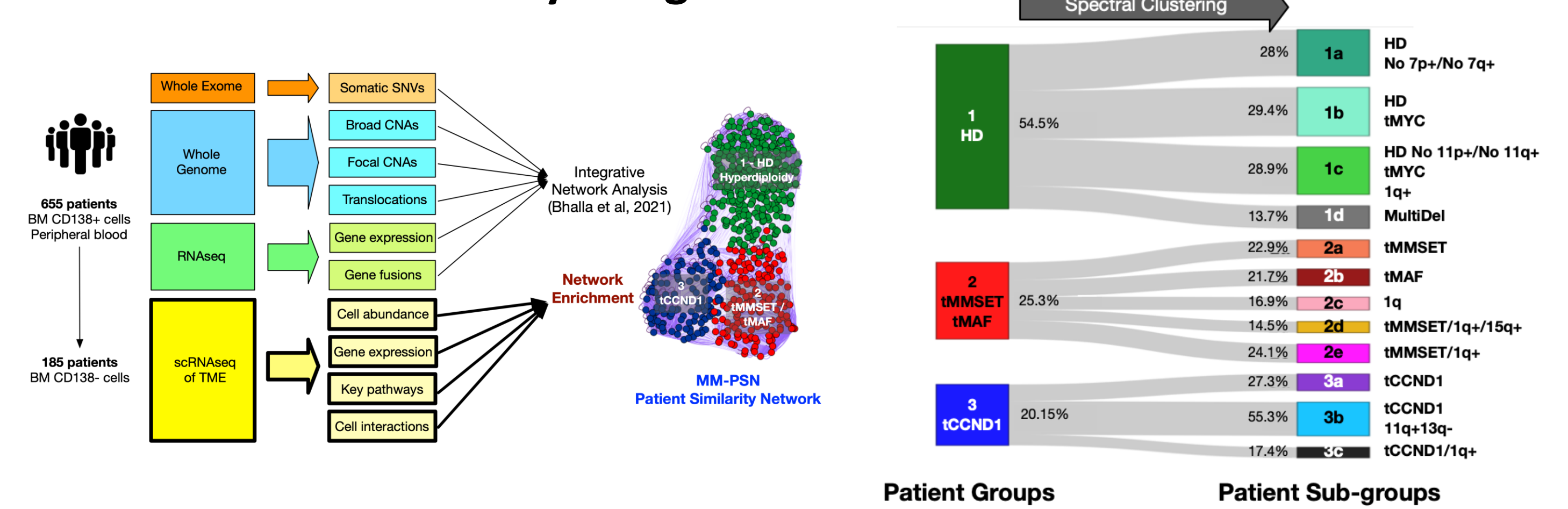
RESULTS

1. Clinical characteristics and Kaplan-Meier curves

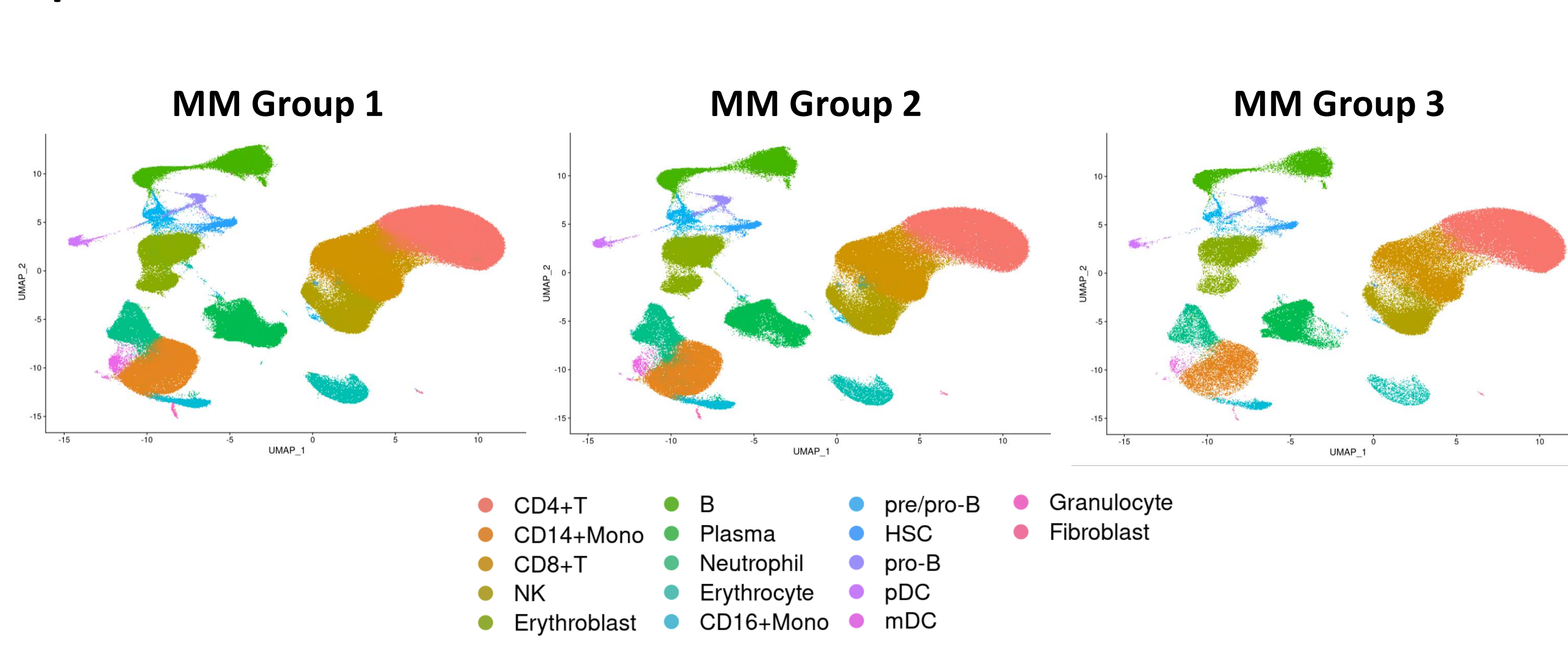
Clinical Variables	Overall	MM Group 1	MM Group 2	MM Group 3
Total number of patients	185	106	46	33
Age (median)	64	63	66.5	62
Gender				
Male	64.32%	60.38%	67.39%	72.73%
Female	35.68%	39.62%	32.61%	27.27%
ISS Stage				
ISS I	30.81%	37.74%	23.91%	18.18%
ISS II	38.92%	38.68%	39.13%	39.39%
ISS III	26.49%	21.70%	28.26%	39.39%
N/A	3.78%	1.89%	8.70%	3.03%



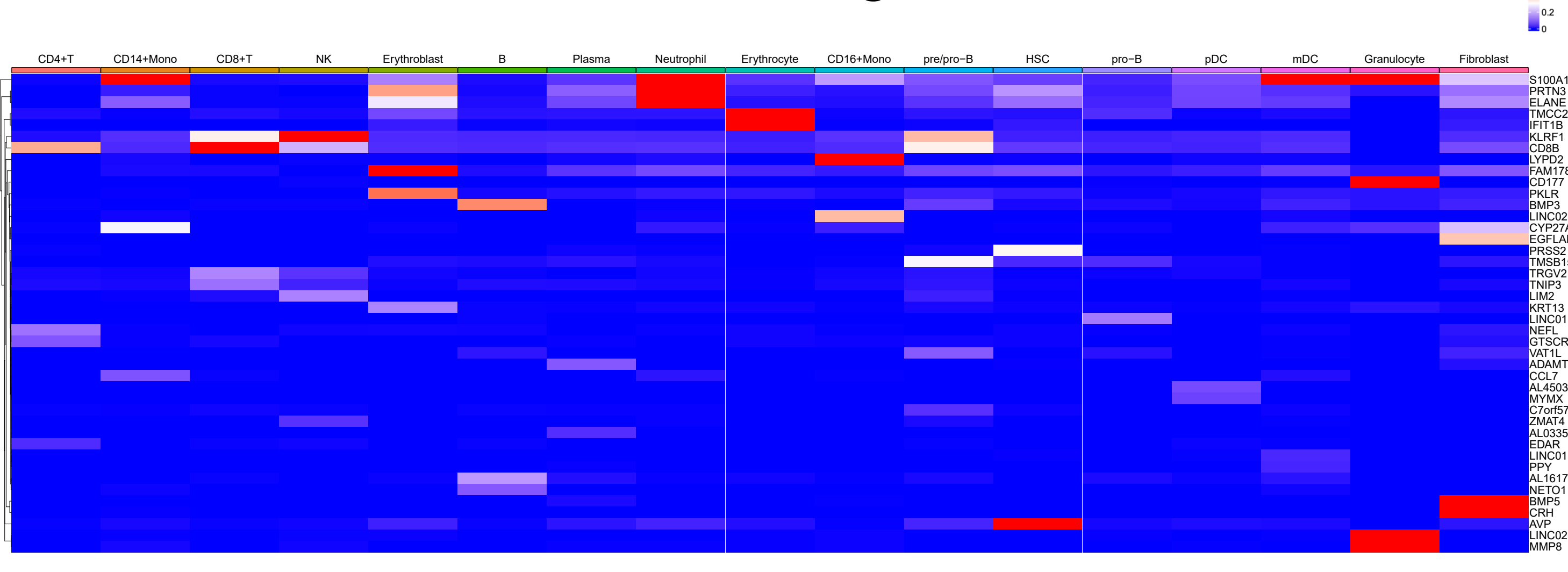
2. Immune MM-PSN study design



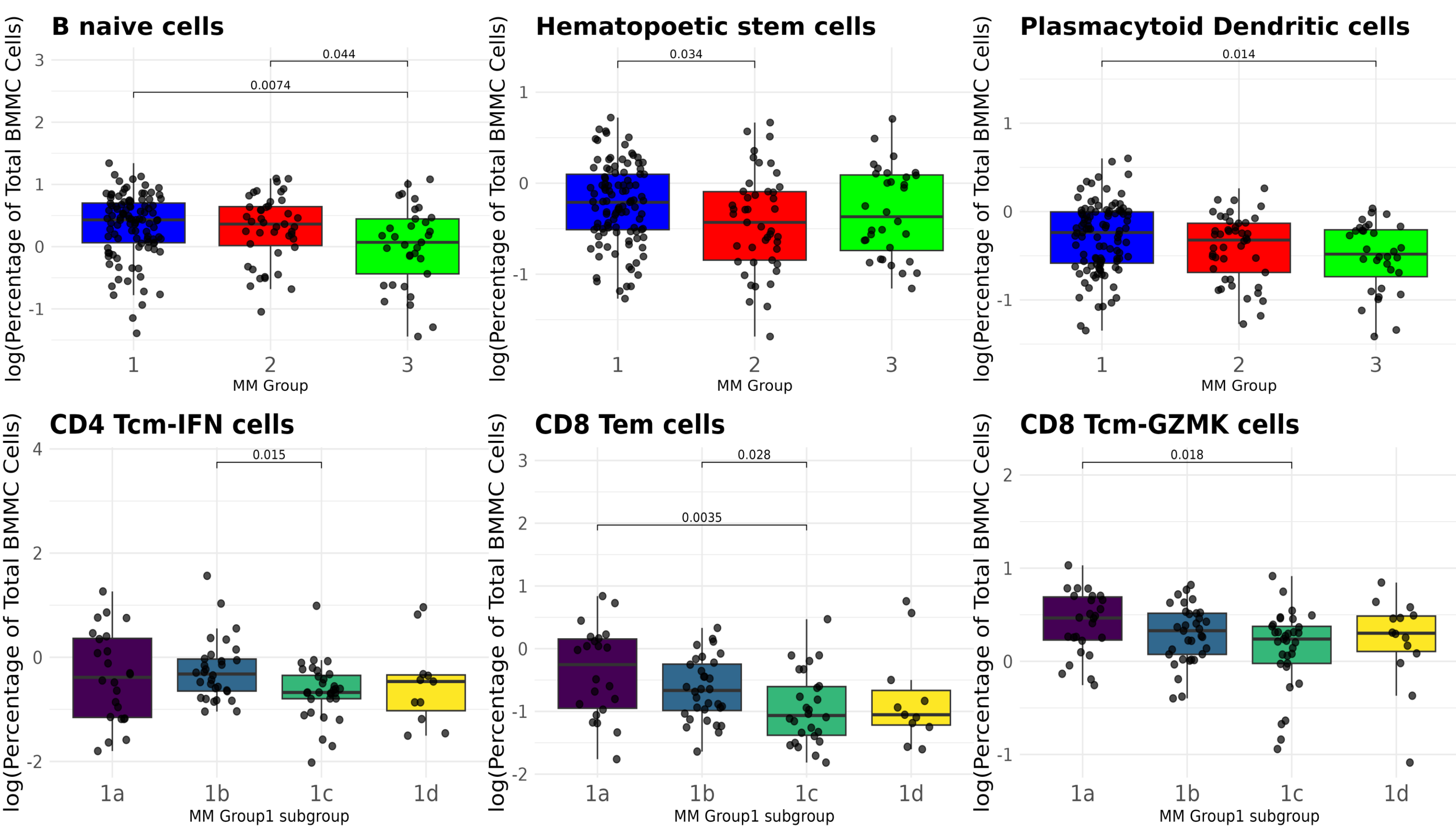
3. Single Cells are Categorized into Distinct Clusters Based on Their Gene Expression Profiles



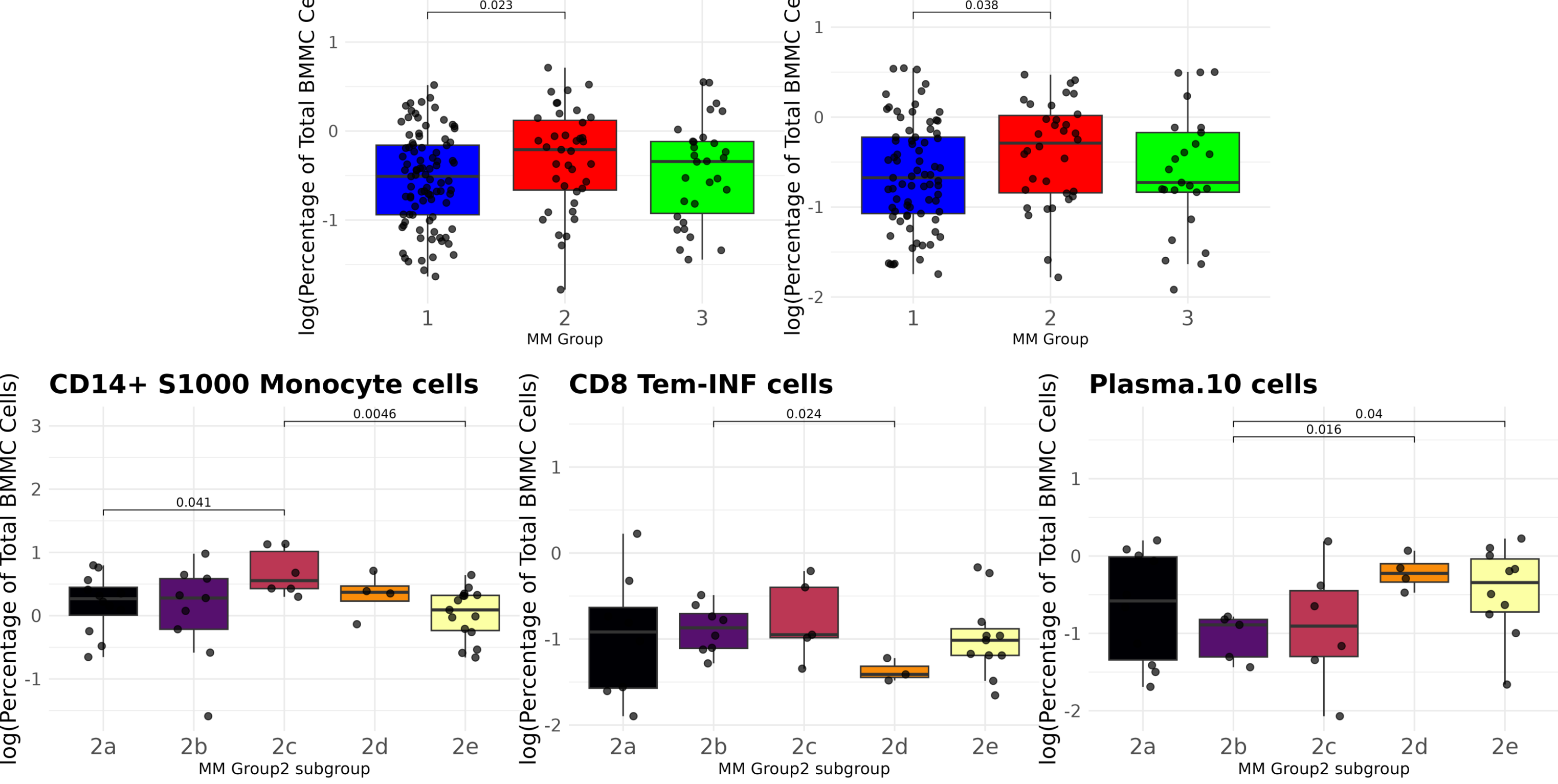
4. Pseudobulk Differential Gene Expression Analysis Reveals Distinct Genes Enriched in each subcluster from our single cell dataset.



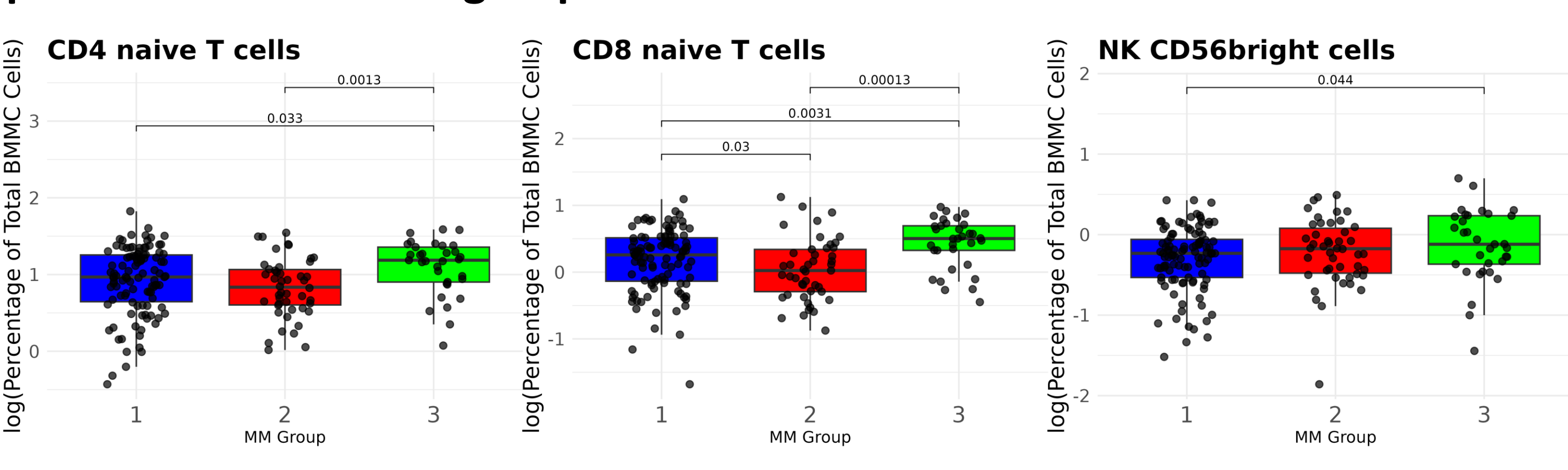
5. MM Group 1 (Hyperdiploidy (HD) and tMYC) revealed a depletion of T cell subsets in the most aggressive hyperdiploid subgroups with gain(1q).



6. MM Group 2 (tMMSET and tMAF) revealed plasma cell expansion and impaired innate immunity characterize aggressive tMMSET subgroups with gain(1q).



7. MM Group 3 (tCCND1) showed enhanced adaptive and innate immune profiles in tCCND1 subgroups correlate with favorable clinical outcomes.



CONCLUSIONS

- Aggressive Subgroups Display Immunosuppressive TMEs:**
- Subgroups with gain(1q), specifically Group MM 1c (tMYC and gain(1q)) and Group MM 2e (tMMSET and gain(1q)), exhibit elevated plasma cells along with decreased CD4 and CD8 T cells. An immunosuppressive TME may facilitate tumor progression, resulting in worse clinical outcomes as seen in gain(1q) subgroups.
- Immune Dysregulation Correlates with Aggressive Disease in tMMSET Subtypes:**
- Group 2 shows expansion of immature B cells and depletion of CD8 T cells, particularly in aggressive subgroups. This immune dysregulation aligns with the aggressive clinical phenotype, emphasizing the role of the TME in disease progression and potential immune evasion mechanisms.
- Favorable Prognosis Linked to Robust Immune Profiles in tCCND1 Subgroups:**
- Group 3 (tCCND1) patients have higher frequencies of naïve CD4 and CD8 T cells and CD56_{high} NK cells, indicating a strong adaptive and innate immune potential. This robust immune landscape correlates with better clinical outcomes, suggesting that these patients might benefit from immunotherapies that leverage their competent immune systems.

CLINICAL RELEVANCE

- Our integrative analysis of bulk tumor genomics and tumor microenvironment (TME) profiling revealed that MM subtypes and their subgroups exhibit distinct immune landscapes correlated with their genetic drivers and clinical outcomes. This comprehensive approach underscores the critical role of the TME in MM pathogenesis and highlights opportunities for improved patient stratification and targeted therapies.
- These findings emphasize that tailoring treatments based on both tumor genetics and TME characteristics holds promise for enhancing therapeutic efficacy and improving prognosis for MM patients. Future research should focus on validating these results in larger cohorts and exploring targeted interventions within the TME to advance personalized medicine in MM.

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